standard quantum limit. A limit, due to the uncertainty principle, on how accurately certain quantities can be measured using standard methods. This limit can be circumvented using quantum nondemolition methods.

stroboscopic measurement. A specific kind of quantum nondemolition measurement in which one makes a sequence of very quick measurements of a vibrating bar, each measurement separated by one vibration period.

structure of a star. The details of how a star's pressure, density, temperature, and gravity change as one goes inward from its surface to its center.

superbomb. A hydrogen bomb that uses a principle by which one can produce an arbitrarily large explosion.

superconductor. A material that conducts electricity perfectly, without any resistance.

supermassive star. A hypothetical star that weighs as much as or more than 10,000 Suns.

supernova. A gigantic explosion of a dying star. The explosion of the star's outer layers is powered by energy that is released when the star's inner core implodes to form a neutron star.

surface gravity. Roughly speaking, the strength of the gravitational pull felt by an observer at rest just above a black hole's horizon. (More precisely: that gravitational pull multiplied by the amount of gravitational time dilation at the observer's location.)

synchrotron radiation. Electromagnetic waves emitted by high-speed electrons that are spiraling around and around magnetic field lines.

thermal pressure. Pressure created by the heat-induced, random motions of atoms, molecules, electrons, and/or other particles.

thermodynamics. The set of physical laws that govern the random, statistical behavior of large numbers of atoms and molecules, including their heat.

thermonuclear reactions. Heat-induced nuclear reactions.

tidal gravity. Gravitational accelerations that squeeze objects along some directions and stretch them along others. Tidal gravity produced by the Moon and Sun is responsible for the tides on the Earth's oceans.

time dilation. A slowing of the flow of time.